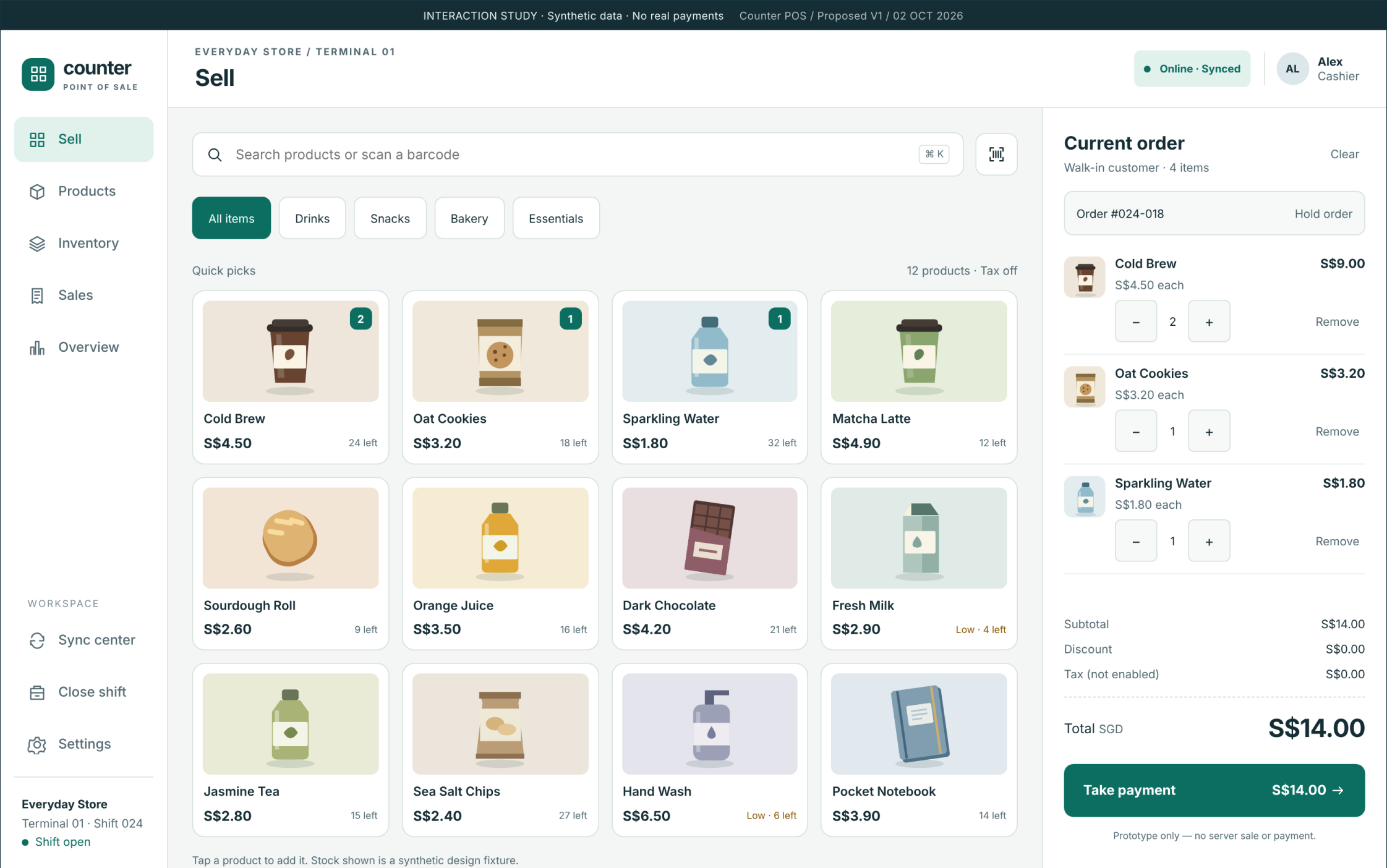


An interface for the whole shift.

UI design / responsive wireframes / engineering diagrams



16
interface studies

08
wireframe boards

04
engineering diagrams

Reading map

Browse interfaces first, then layout intent and transaction boundaries.

03	Sell / desktop
04	Sell / tablet landscape
05	Sell to payment / mobile
06	Cash payment / desktop
07	Receipt / desktop
08	Products / back office
09	Inventory / ledger and adjustment
10	Sales history / back office
11	Daily overview
12	Sync center / pending cash sales
13	Shift reconciliation
14	Settings / update and recovery
15	Workspace entry / design demo
16	Offline selling / mobile

17	Wireframe / 01 desktop pos
18	Wireframe / 02 tablet pos
19	Wireframe / 03 mobile flow
20	Wireframe / 04 payment
21	Wireframe / 05 products
22	Wireframe / 06 inventory
23	Wireframe / 07 sales return
24	Wireframe / 08 sync shift
25	Engineering diagram / 01 architecture
26	Engineering diagram / 02 core erd
27	Engineering diagram / 03 sale sync sequence
28	Engineering diagram / 04 workflow states

Sell / desktop

Product discovery and persistent cart. Default basket: S\$14.00, tax disabled.

counter
POINT OF SALE

Sell

Products

Inventory

Sales

Overview

WORKSPACE

Sync center

Close shift

Settings

Everyday Store
Terminal 01 · Shift 024
● Shift open

EVERYDAY STORE / TERMINAL 01

Sell

Search products or scan a barcode

All itemsDrinksSnacksBakeryEssentials

Quick picks

12 products · Tax off



2

Cold Brew
S\$4.50
24 left



1

Oat Cookies
S\$3.20
18 left



1

Sparkling Water
S\$1.80
32 left



Matcha Latte
S\$4.90
12 left



Sourdough Roll
S\$2.60
9 left



Orange Juice
S\$3.50
16 left



Dark Chocolate
S\$4.20
21 left



Fresh Milk
S\$2.90
Low · 4 left



Jasmine Tea
S\$2.80
15 left



Sea Salt Chips
S\$2.40
27 left



Hand Wash
S\$6.50
Low · 6 left



Pocket Notebook
S\$3.90
14 left

Online · Synced

AL Alex
Cashier

Current order

Walk-in customer · 4 items

Order #024-018

Hold order



Cold Brew
S\$4.50 each

- 2 +

Remove



Oat Cookies
S\$3.20 each

- 1 +

Remove



Sparkling Water
S\$1.80 each

- 1 +

Remove

Subtotal

S\$14.00

Discount

S\$0.00

Tax (not enabled)

S\$0.00

Total SGD

S\$14.00

Take paymentS\$14.00 →

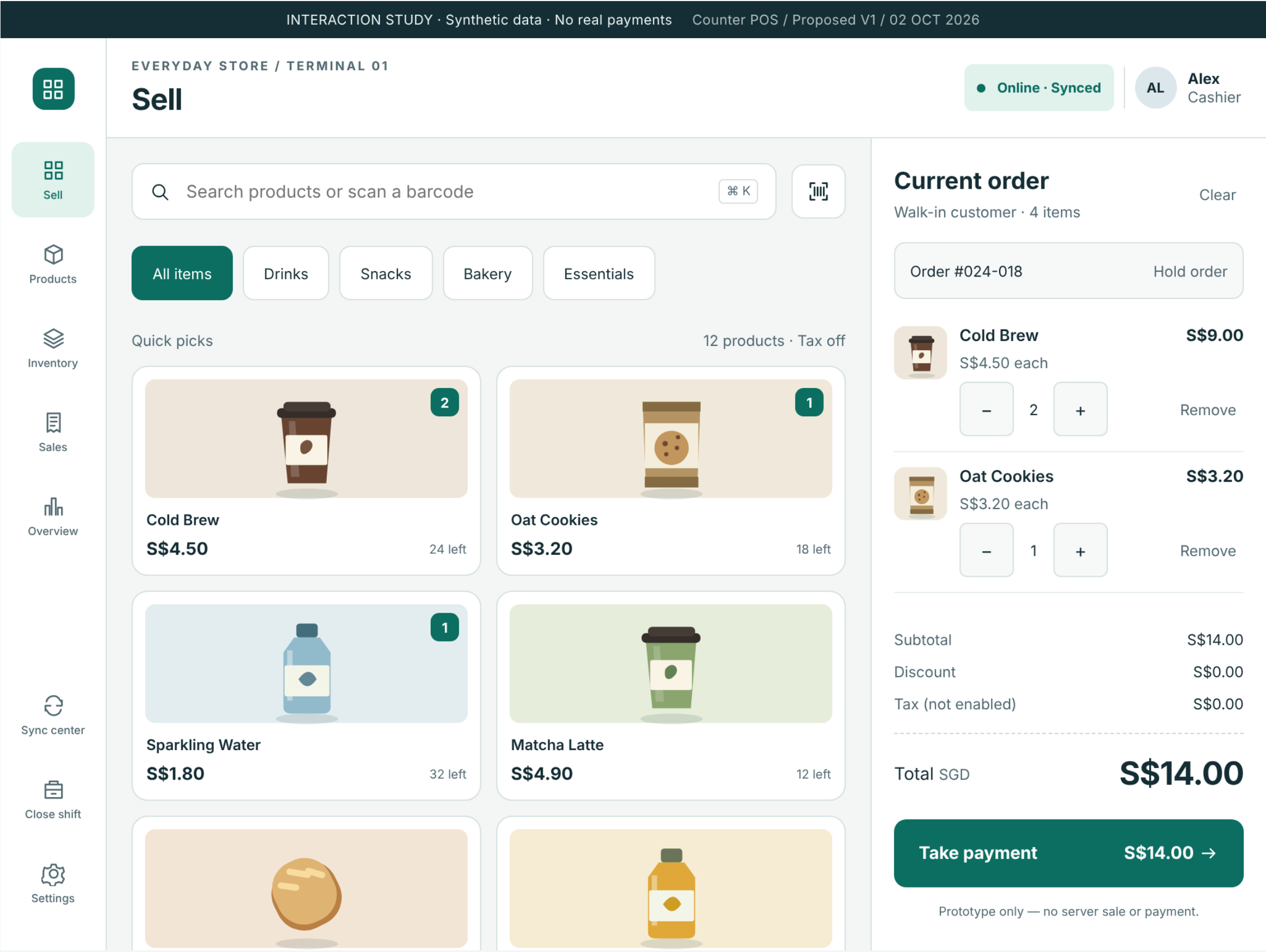
Prototype only — no server sale or payment.

Proposed V1 | Synthetic data | Working identity | 02 October 2026

3

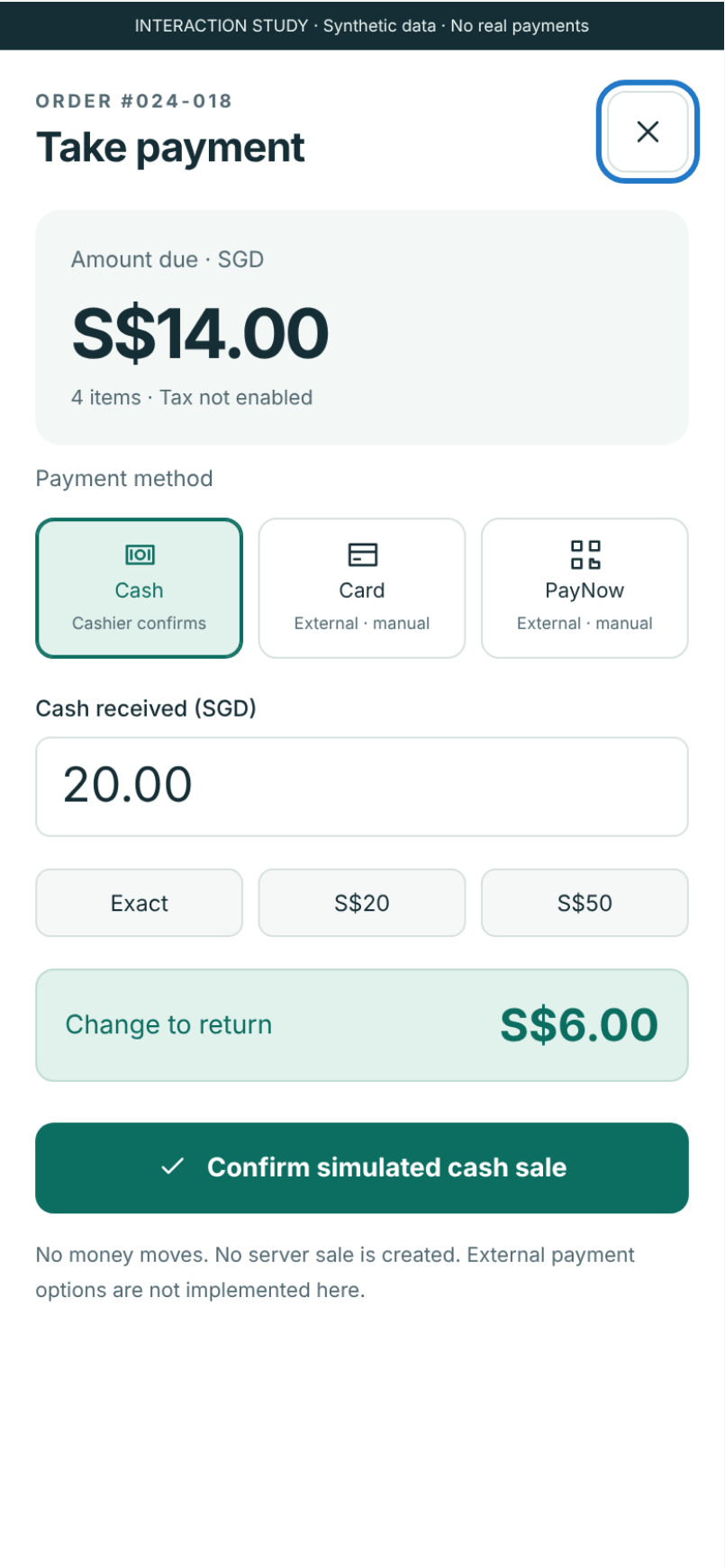
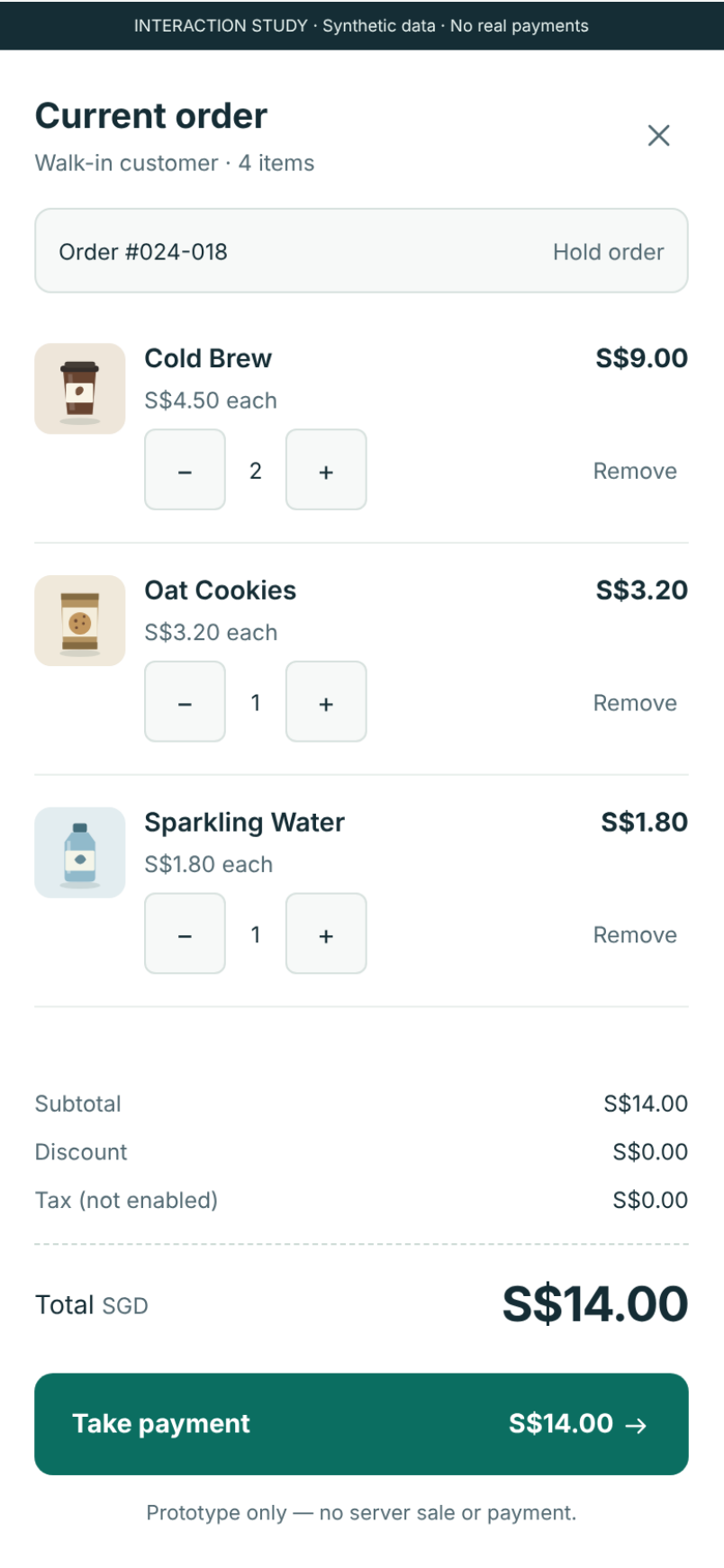
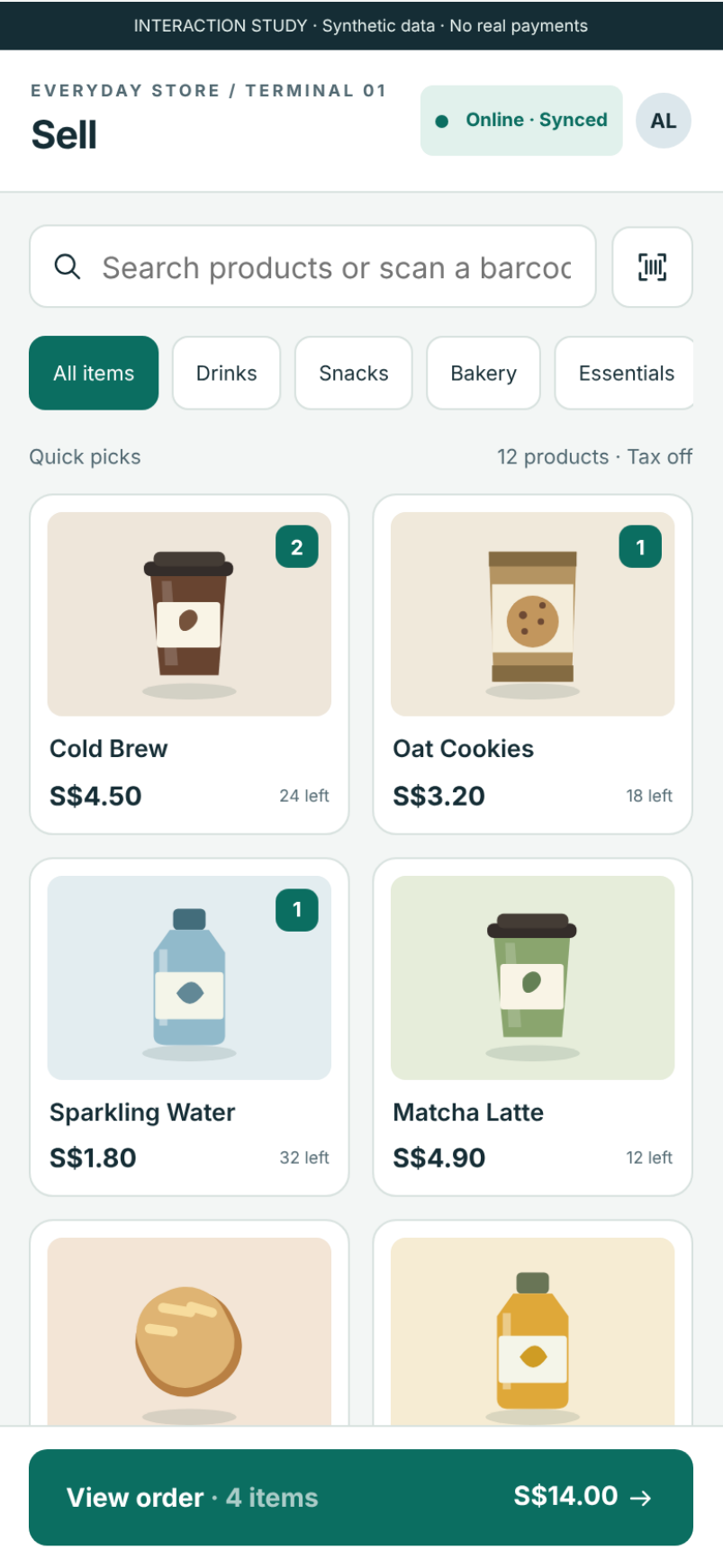
Sell / tablet landscape

Compact navigation, touch-friendly products and a readable cart.



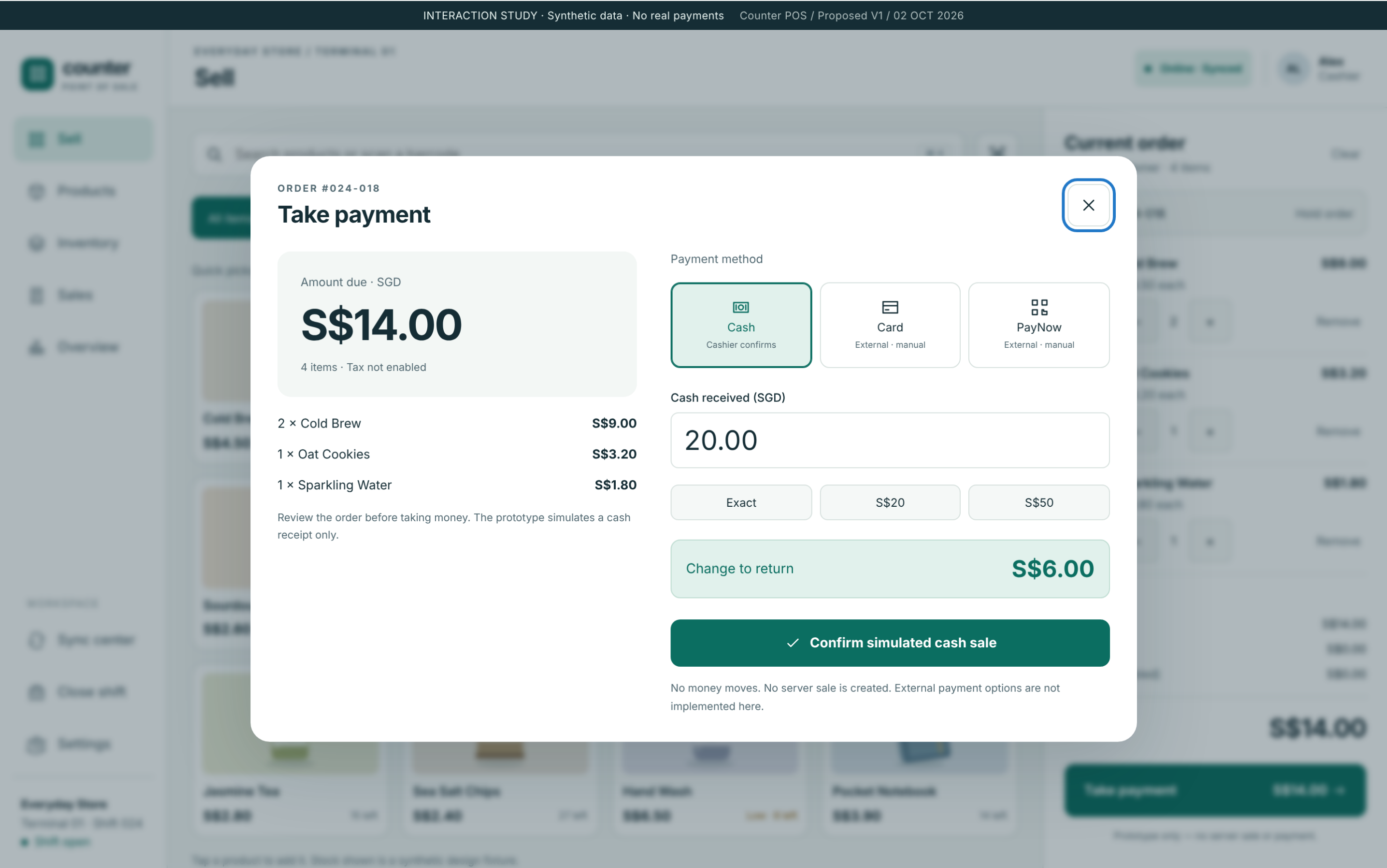
Sell to payment / mobile

Three focused steps at 390 x 844 CSS pixels. No real payments or persistence.



Cash payment / desktop

Cash received S\$20.00; change S\$6.00. External methods are not gateway integrations.



Receipt / desktop

Simulated receipt. Print and reprint do not create another sale.

INTERACTION STUDY · Synthetic data · No real payments

Counter POS / Proposed V1 / 02 OCT 2026

counter

POINT OF SALE

Sell

Products

Inventory

Sales

Overview

Sync center

Close shift

Settings

Everyday Store

Terminal 01 · Shift 001

Shift open

EVERYDAY STORE / TERMINAL 01

Sell

Back

Print order

2 × Cold Brew

1 × Oat Cookies

1 × Sparkling Water

Subtotal

Tax (off)

TOTAL SGD

Cash received

Change

Thank you. See you again.

SYNTHETIC · NOT A TAX INVOICE

SS\$14.00

SIMULATED RECEIPT

×

✓

Sale preview complete

Cash received S\$20.00.

Return **S\$6.00** in change.

Demo only · No server record

Start a new order →

Print preview

Printing or reprinting must never create another sale.

This receipt is sample data, not a tax invoice.

EVERYDAY STORE

COUNTER POS · DEMO

02 OCT 2026 · 14:32

CP-000018

2 × Cold Brew9.00

1 × Oat Cookies3.20

1 × Sparkling Water1.80

Subtotal14.00

Tax (off)0.00

TOTAL SGD14.00

Cash received20.00

Change6.00

Thank you. See you again.

SYNTHETIC · NOT A TAX INVOICE

Proposed V1 | Synthetic data | Working identity | 02 October 2026

7

Products / back office

Catalogue, unique identifiers, current prices and active/low-stock states.

INTERACTION STUDY · Synthetic data · No real payments

Counter POS / Proposed V1 / 02 OCT 2026

counter
POINT OF SALE

Sell

Products

Inventory

Sales

Overview

WORKSPACE

Sync center

Close shift

Settings

Everyday Store

Terminal 01 · Shift 024

● Shift open

EVERYDAY STORE / TERMINAL 01

Products

● Online · Synced

MG Morgan
Manager

Your product catalogue

Simple products. Clear prices. Stock lives in the ledger.

+ Add product

Active products

12

Across 4 categories

Low-stock items

2

Fresh Milk · Hand Wash

Price revisions

16

History is preserved

Catalogue status

Current

Synthetic snapshot v12

Search name, SKU or barcode

All categories

Active

PRODUCT	CATEGORY	SELLING PRICE	ON HAND	STATUS	
 Cold Brew DRK-001	Drinks	S\$4.50	24	Active	Edit →
 Oat Cookies SNK-001	Snacks	S\$3.20	18	Active	Edit →
 Sparkling Water DRK-002	Drinks	S\$1.80	32	Active	Edit →
 Matcha Latte DRK-003	Drinks	S\$4.90	12	Active	Edit →
 Sourdough Roll FD-001	Bakery	S\$2.60	9	Active	Edit →
 Orange Juice DRK-004	Drinks	S\$3.50	16	Active	Edit →

Inventory / ledger and adjustment

A counted quantity becomes a reasoned movement, not a direct overwrite.

counter
POINT OF SALE

Sell

Products

Inventory

Sales

Overview

WORKSPACE

Sync center

Close shift

Settings

Everyday Store
Terminal 01 · Shift 024
● Shift open

INTERACTION STUDY · Synthetic data · No real paymentsCounter POS / Proposed V1 / 02 OCT 2026

EVERYDAY STORE / TERMINAL 01

Inventory

Online · Synced

MG Morgan
Manager

Inventory ledger

Every quantity has a reason and a reference.

Units on hand

198

Current synthetic balance

Low-stock items

2

Reorder attention needed

Stock received

60

Today · sample ledger

Adjustments

1

Manager recorded

Recent movements

PRODUCT	MOVEMENT	QUANTITY	REFERENCE
 Cold Brew DRK-001	Sale	-2	CP-000018
 Oat Cookies SNK-001	Sale	-1	CP-000018
 Sparkling Water DRK-002	Sale	-1	CP-000018
 Fresh Milk DRK-005	Adjustment	-1	ADJ-0003
 Matcha Latte DRK-003	Receipt	+20	RCV-0007

Count an item

Product

Fresh Milk · DRK-005

Recorded stock

4

Counted stock

3

Adjustment preview: -1 unit

Reason

Damaged pack

A reason and current balance version are required.

Review adjustment

Manager only · Online required
Creates a movement, not a direct stock overwrite.

Sales history / back office

Immutable posted sales with linked returns; manual methods are labeled.

counter
POINT OF SALE

Sell

Products

Inventory

Sales

Overview

WORKSPACE

Sync center

Close shift

Settings

Everyday Store
Terminal 01 · Shift 024
● Shift open

EVERYDAY STORE / TERMINAL 01

Online · Synced

MG Morgan
Manager

Sales history

Export CSV

Gross sales

\$S\$486.00

Before refunds

Refunds

\$S\$18.00

Recognized today

Net sales

\$S\$468.00

Gross less refunds

Completed orders

36

Synthetic daily data

Find a receipt

02 Oct 2026

All methods

RECEIPT	TIME	METHOD	ITEMS	TOTAL	STATE	
CP-000018	14:32	Cash	4	S\$14.00	Posted	View →
CP-000017	14:26	Card · manual	3	S\$12.90	Posted	View →
CP-000016	14:18	Cash	2	S\$7.70	Posted	View →
CP-000015	14:05	PayNow · manual	5	S\$19.60	Posted	View →
CP-000014	13:54	Cash	3	S\$11.20	Part returned	View →
CP-000013	13:41	Cash	1	S\$4.50	Posted	View →

Manual methods are operator-recorded external payments, not provider-verified transactions. Listed rows are a subset of the synthetic daily summary.

Daily overview

Synthetic daily gross S\$486.00, refunds S\$18.00, net S\$468.00.

counter
POINT OF SALE

Sell

Products

Inventory

Sales

Overview

Workspace

Sync center

Close shift

Settings

Everyday Store
Terminal 01 · Shift 024
● Shift open

INTERACTION STUDY · Synthetic data · No real payments Counter POS / Proposed V1 / 02 OCT 2026

EVERYDAY STORE / TERMINAL 01

Online · Synced MG Morgan Manager

Overview

A clear view of today

02 October 2026 · Server-confirmed synthetic sales only

Today ▼

Gross sales

S\$486.00

36 completed orders

Refunds

S\$18.00

2 return records

Net sales

S\$468.00

Gross less refunds

Average ticket

S\$13.50

Gross / completed orders

Sales through the day

Illustrative hourly shape

9:00

10:00

11:00

12:00

13:00

14:00

15:00

16:00

Illustrative design visualization; not an analytic dataset.

Most purchased

01 Cold Brew22 units

02 Sparkling Water18 units

03 Oat Cookies14 units

04 Orange Juice11 units

Needs attention

2 products are running low

Fresh Milk: 4 · Hand Wash: 6

All sample sales synced

Pending local amounts must be shown separately from confirmed reports.

Sync center / pending cash sales

Pending local sales are separate from server-confirmed totals.

INTERACTION STUDY · Synthetic data · No real payments

Counter POS / Proposed V1 / 02 OCT 2026

counter

POINT OF SALE

Sell

Products

Inventory

Sales

Overview

WORKSPACE

Sync center

Close shift

Settings

Everyday Store

Terminal 01 · Shift 024

● Shift open

EVERYDAY STORE / TERMINAL 01

Sync center

● Offline · 3 pending

MG Morgan Manager

See what is waiting to sync

Review what is saved locally and what the server has acknowledged.

Offline simulation · 3 sales waiting

Keep this device's data. Reopen the app when connected to complete foreground sync.

Pending sales

3

Saved on this device

Pending amount

S\$32.20

Not in confirmed totals

Needs review

0

No sample conflicts

Last server sync

14:24

Sample timestamp

Pending documents

LOCAL-T01-018	S\$14.00 · Cash	Saved 14:32	Pending
LOCAL-T01-019	S\$7.70 · Cash	Saved 14:35	Pending
LOCAL-T01-020	S\$10.50 · Cash	Saved 14:38	Pending

Retry reuses the same transaction UUID. An acknowledgement never creates a second stock deduction.

Safe recovery

Proposed V1 | Synthetic data | Working identity | 02 October 2026

12

Shift reconciliation

S\$100 float + S\$286 cash sales - S\$18 refunds - S\$20 cash out = S\$348.

INTERACTION STUDY · Synthetic data · No real payments

Counter POS / Proposed V1 / 02 OCT 2026

counter

POINT OF SALE

Sell

Products

Inventory

Sales

Overview

WORKSPACE

Sync center

Close shift

Settings

Everyday Store

Terminal 01 · Shift 024

● Shift open

EVERYDAY STORE / TERMINAL 01

Close shift

● Online · Synced

AL Alex
Cashier

Close shift 024

Terminal 01 · Alex · Opened at 09:00

Ready to reconcile

Expected cash

Opening float

S\$100.00

Cash sales applied

S\$286.00

Cash refunds

−S\$18.00

Cash in

S\$0.00

Cash out

−S\$20.00

Expected drawer

S\$348.00

Change is already excluded from cash applied.

This shift fixture is consistent with the daily totals.

Count the drawer

Actual cash count (SGD)

348.00

Difference

S\$0.00

✓ No pending or review documents

✓ Server reconciliation acknowledged

Confirm close shift

Online required. A local cash-count draft is not a closed shift.

Design study · Synthetic data · Administrative controls are read-only in this prototype.

Proposed V1 | Synthetic data | Working identity | 02 October 2026

13

Settings / update and recovery

A waiting update requires explicit, safe activation. This study has no service worker.

INTERACTION STUDY · Synthetic data · No real payments

Counter POS / Proposed V1 / 02 OCT 2026

 **counter**
POINT OF SALE

Sell

Products

Inventory

Sales

Overview

WORKSPACE

Sync center

Close shift

Settings

Everyday Store

Terminal 01 · Shift 024

● Shift open

EVERYDAY STORE / TERMINAL 01

Settings

Online · Synced

MG Morgan Manager

Store & device

Small configuration surface. Important changes stay auditable.

Store preferences

Store

Everyday Store

Review

Currency

SGD · Locked after first sale

Review

Timezone

Asia/Singapore

Review

Tax

Disabled · Demo configuration

Review

Receipt

80mm browser print

Review

App lifecycle

Design build 1.0

New version available

Activate only when checkout and outbox are safe.

Review update

Prototype illustration only. No service worker is registered.

Backup & recovery

Server backups and pending-device recovery are separate responsibilities. Test a restore before pilot use.

View recovery guide

Proposed V1 | Synthetic data | Working identity | 02 October 2026

14

Workspace entry / design demo

No credentials are collected. Production enrollment and sign-in are future work.

INTERACTION STUDY · Synthetic data · No real authentication

 **counter**
POINT OF SALE

A CALMER WAY TO SELL

A small counter.
A complete day.

Products, sales and stock, together.
A focused workspace for everyday retail.



Single-store focus · Cash-first offline design

COUNTER POS / EVERYDAY STORE

Open your workspace

In the proposed application, sign in online and verify this terminal before opening a shift.

Workspace

Everyday Store — demo

Operator

Alex — synthetic cashier

Opening float (SGD)

100.00

Enter design demo →

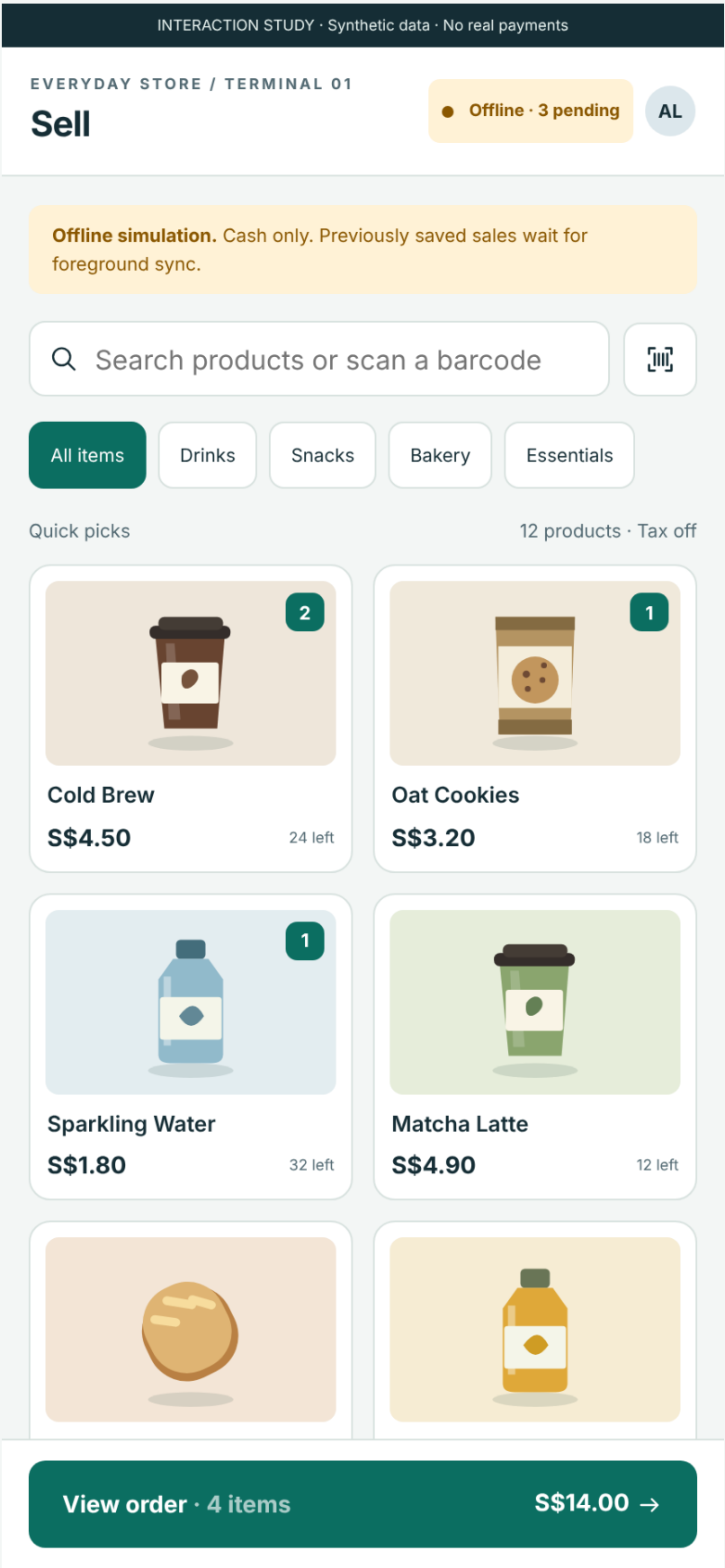
No account or password is collected.
This study is not a deployed POS.

Proposed V1 | Synthetic data | Working identity | 02 October 2026

15

Offline selling / mobile

430 x 932 CSS pixels. Amber status communicates bounded offline cash-selling design.



Wireframe / 01 desktop pos

Editable SVG source and matching PNG preview included. Hierarchy is authoritative; this is not a deployed UI.

COUNTER POS / WF-01

Desktop selling workspace

Product discovery and order review remain visible together.

COUNTER POS

Everyday Store / Terminal 01

Online / Cashier

Sell

Products

Inventory

Sales

Overview

1

Search product name, SKU or barcode

All items

Drinks

Snacks

Bakery

2

Product name
S\$0.00

Product name
S\$0.00

Product name
S\$0.00

Product name
S\$0.00

Product name
S\$0.00

Product name
S\$0.00

Product name
S\$0.00

Product name
S\$0.00

Current order

Walk-in / 4 items

Product × quantity

Unit price / line total

- 1 +

Product × quantity

Unit price / line total

- 1 +

Product × quantity

Unit price / line total

- 1 +

Subtotal / discount / tax

TOTAL

S\$14.00

3

Take payment →

1 Always-available scan/search 2 Flexible, scrollable product grid 3 Fixed total and payment action

Proposed V1 · Design contract, not a deployed application · 02 October 2026

Proposed V1 | Synthetic data | Working identity | 02 October 2026

17

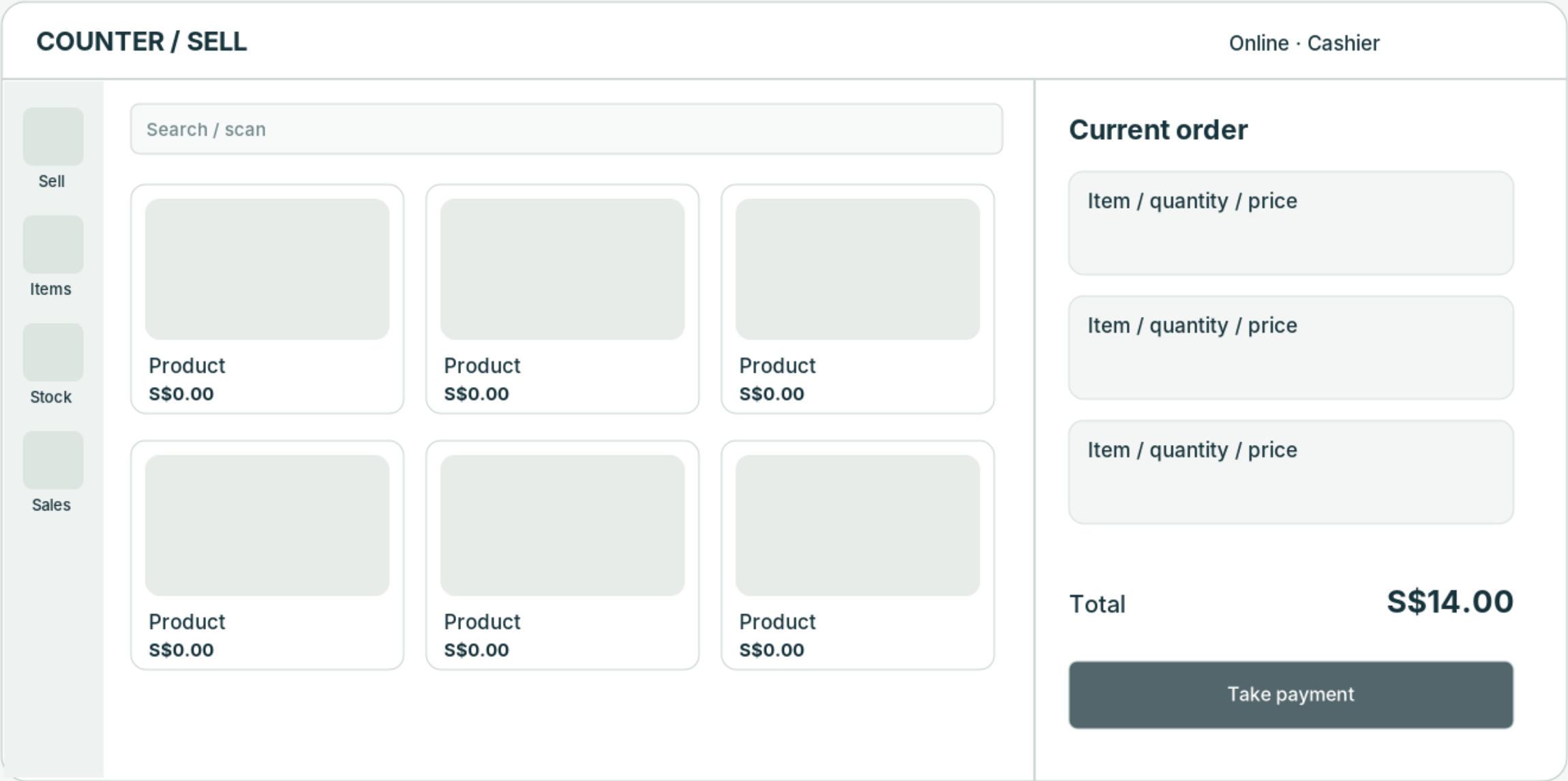
Wireframe / 02 tablet pos

Editable SVG source and matching PNG preview included. Hierarchy is authoritative; this is not a deployed UI.

COUNTER POS / WF-02

Tablet selling workspace

Compact navigation; keep a readable order panel in landscape.



Keep at least 44px action targets. At narrower portrait widths, switch to mobile order navigation.

Proposed V1 · Design contract, not a deployed application · 02 October 2026

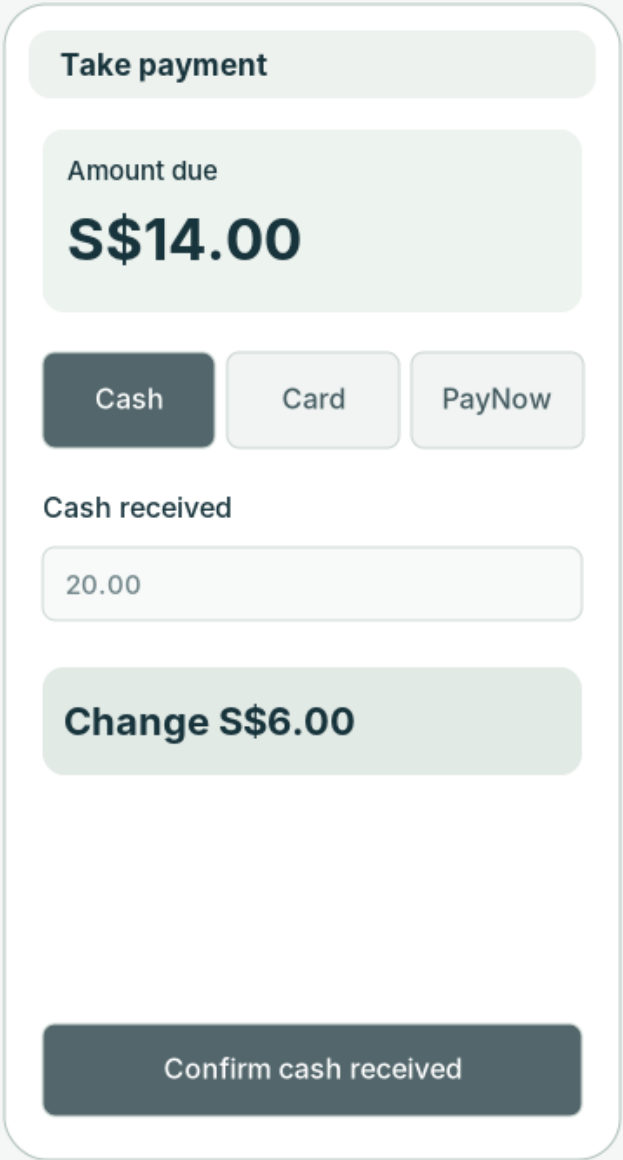
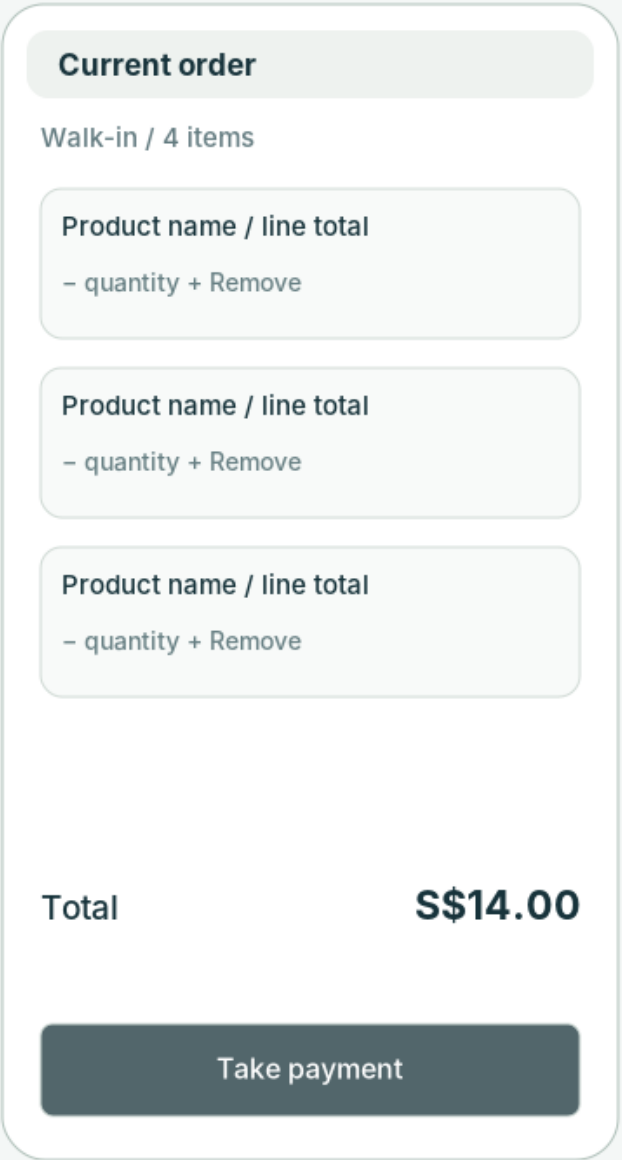
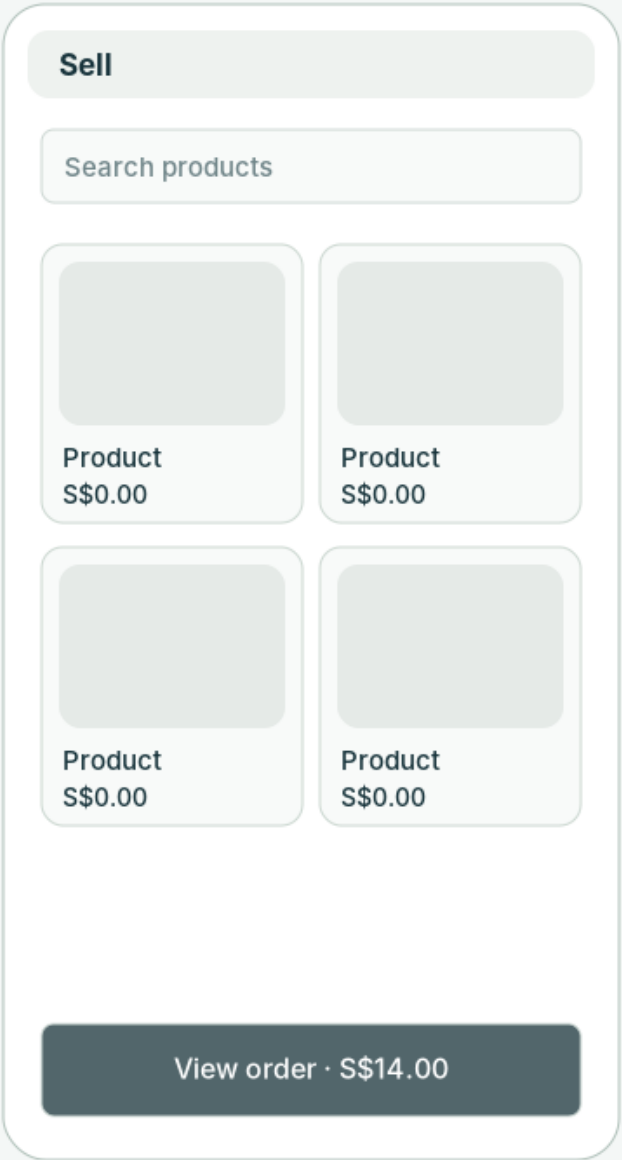
Wireframe / 03 mobile flow

Editable SVG source and matching PNG preview included. Hierarchy is authoritative; this is not a deployed UI.

COUNTER POS / WF-03

Mobile selling, order and payment

Separate focused screens replace a cramped desktop cart.



Safe-area-aware bottom action. Product grid and line list scroll independently. Do not disable browser zoom.

Wireframe / 04 payment

Editable SVG source and matching PNG preview included. Hierarchy is authoritative; this is not a deployed UI.

COUNTER POS / WF-04

Payment review and completion

Explicit money state, one selected method and a visible change amount.

Take payment

Amount due

\$S\$14.00

2 × Cold Brew 9.00

1 × Oat Cookies 3.20

1 × Sparkling Water 1.80

Cash

Card

PayNow

Cash received (SGD)

20.00

Change \$S\$6.00

Confirm cash received

Disable while saving; show success only after the local transaction completes.

Card / PayNow are manual external records online only. A selected method is not provider verification.

Wireframe / 05 products

Editable SVG source and matching PNG preview included. Hierarchy is authoritative; this is not a deployed UI.

COUNTER POS / WF-05

Products and the product editor

Catalogue maintenance is separate from receiving stock.

COUNTER POS

Everyday Store / Terminal 01

Online / Cashier

Sell

Products

Inventory

Sales

Overview

Products

Search / filter category / active

Product / SKU	Category	Price	Stock	Status
Product name / ITEM-001	Category	S\$0.00	24	Active
Product name / ITEM-002	Category	S\$0.00	24	Active
Product name / ITEM-003	Category	S\$0.00	24	Active
Product name / ITEM-004	Category	S\$0.00	24	Active
Product name / ITEM-005	Category	S\$0.00	24	Active
Product name / ITEM-006	Category	S\$0.00	24	Active

Edit product

Name

Value

SKU / barcode

Value

Category

Value

Selling price / cost

Value

Low-stock threshold

Value

Save product

A product is not stock. Save validates uniqueness; receiving inventory creates a separate ledger event.

Wireframe / 06 inventory

Editable SVG source and matching PNG preview included. Hierarchy is authoritative; this is not a deployed UI.

COUNTER POS / WF-06

Inventory movements and counted adjustment

A reason and an expected version prevent silent overwrites.

COUNTER POS

Everyday Store / Terminal 01

Online / Cashier

Sell

Products

Inventory

Sales

Overview

Inventory ledger

Units on hand

198

Low-stock items

2

Open exceptions

0

Product / Type / Signed quantity / Reference

Cold Brew SALE -2 CP-000018

Oat Cookies SALE -1 CP-000018

Fresh Milk ADJUST -1 ADJ-0003

Matcha Latte RECEIVE +20 RCV-0007

Hand Wash RETURN +1 REF-0002

Count an item

Product

Value

Recorded / counted stock

Value

Reason

Value

Preview: -1 unit

Review adjustment

Server transaction appends a movement and updates the balance projection together. Manager-only, online.

Wireframe / 07 sales return

Editable SVG source and matching PNG preview included. Hierarchy is authoritative; this is not a deployed UI.

COUNTER POS / WF-07

Sales detail and return authorization

Returns reference original immutable lines; no completed-sale delete.

COUNTER POS

Everyday Store / Terminal 01

Online / Cashier

Sell

Products

Inventory

Sales

Overview

Sales history

Receipt / date / payment method

CP-000018 14:32 Cash S\$14.00 Posted

CP-000017 14:31 Cash S\$14.00 Posted

CP-000016 14:30 Cash S\$14.00 Posted

CP-000015 14:29 Cash S\$14.00 Posted

CP-000014 14:28 Cash S\$14.00 Posted

CP-000013 14:27 Cash S\$14.00 Posted

Return items

Original receipt CP-000018

Item / remaining qty / return qty

Restock: yes / no

Item / remaining qty / return qty

Restock: yes / no

Item / remaining qty / return qty

Restock: yes / no

Reason required · Manager only

Refund total S\$3.20

Review refund

Lock original lines; cap cumulative refund quantity and cents; restock only if physically resalable.

Wireframe / 08 sync shift

Editable SVG source and matching PNG preview included. Hierarchy is authoritative; this is not a deployed UI.

COUNTER POS / WF-08

Sync recovery and shift-close guard

Expose local state, retry safely and protect unresolved documents.

COUNTER POS

Everyday Store / Terminal 01

Online / Cashier

Sell

Products

Inventory

Sales

Overview

Offline: 3 transactions saved on this device

Reopen the app when connected to complete foreground synchronization.

LOCAL-T01-018 S\$14.00 CASH

State: PENDING / same UUID on every retry

LOCAL-T01-019 S\$14.00 CASH

State: PENDING / same UUID on every retry

LOCAL-T01-020 S\$14.00 CASH

State: PENDING / same UUID on every retry

Retry sync

Manager recovery export

Never offer “clear browser data” as a recovery action. Never label a pending sale as server-synced.

Close shift

Expected cash / counted cash

Variance / required reason

Pending or review sales: 3

Reconciliation: not ready

Final close blocked

AUTH_REQUIRED, RETRY and NEEDS_REVIEW need distinct actions. Background Sync is optional, not assumed.

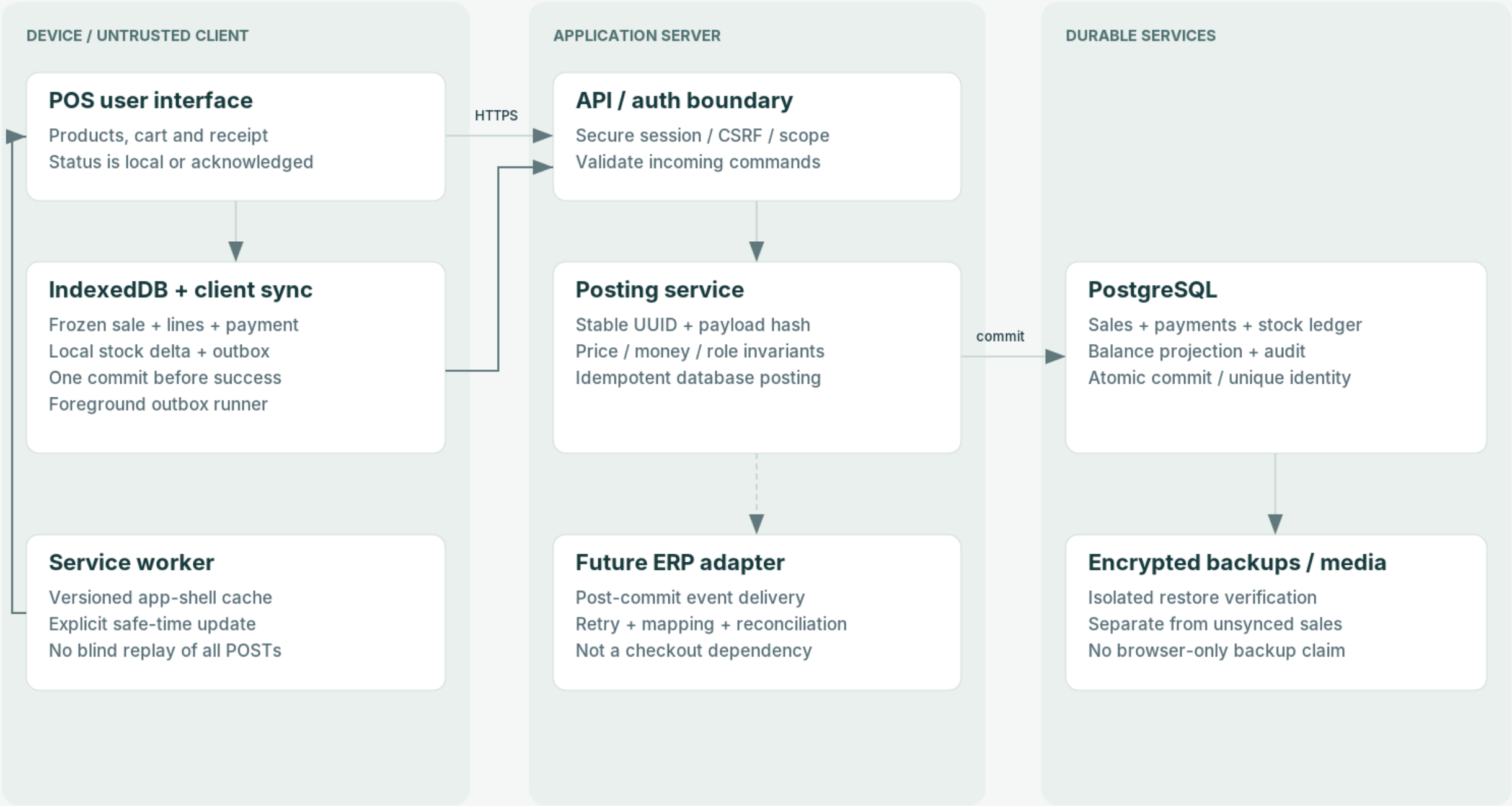
Engineering diagram / 01 architecture

Editable SVG and Mermaid source included. Read the full contract for exact constraints.

COUNTER POS / ARCH-01

A modular POS, with explicit trust boundaries

PWA client + local outbox + server transaction. ERP remains optional.



Design decision · Same-origin client/API · Browser APIs and lifecycle sources: S02–S06

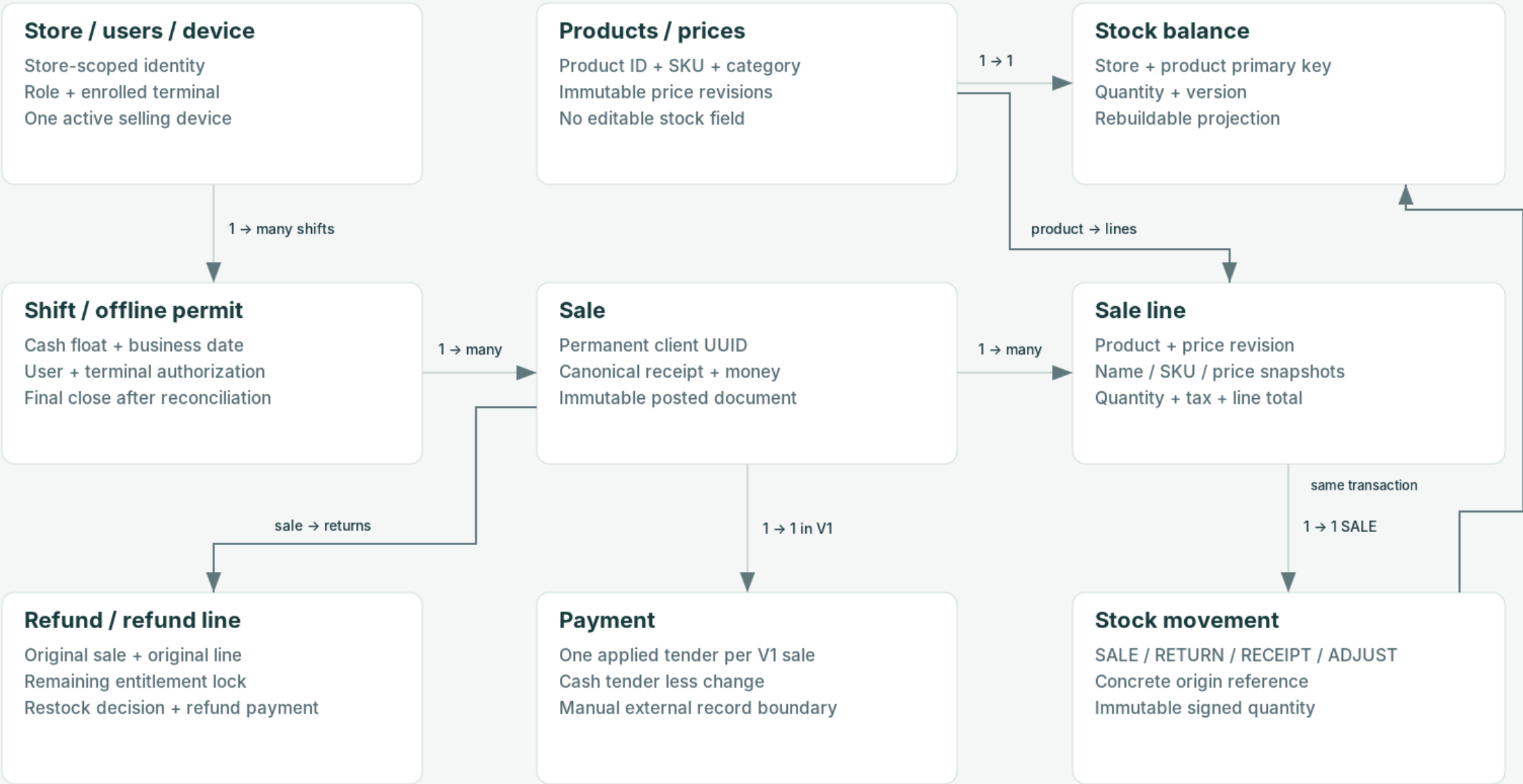
Engineering diagram / 02 core erd

Editable SVG and Mermaid source included. Read the full contract for exact constraints.

COUNTER POS / DATA-01

Core entity relationships

Conceptual reading view. The SQL contract contains all tables, keys and invariants.



Ledger is authoritative; balance is a projection. All references remain store-scoped.
Refunds never delete sales. Exact compound keys, constraints and omitted support tables: specs/schema.sql.

Proposed V1 · Design contract, not a deployed application · 02 October 2026

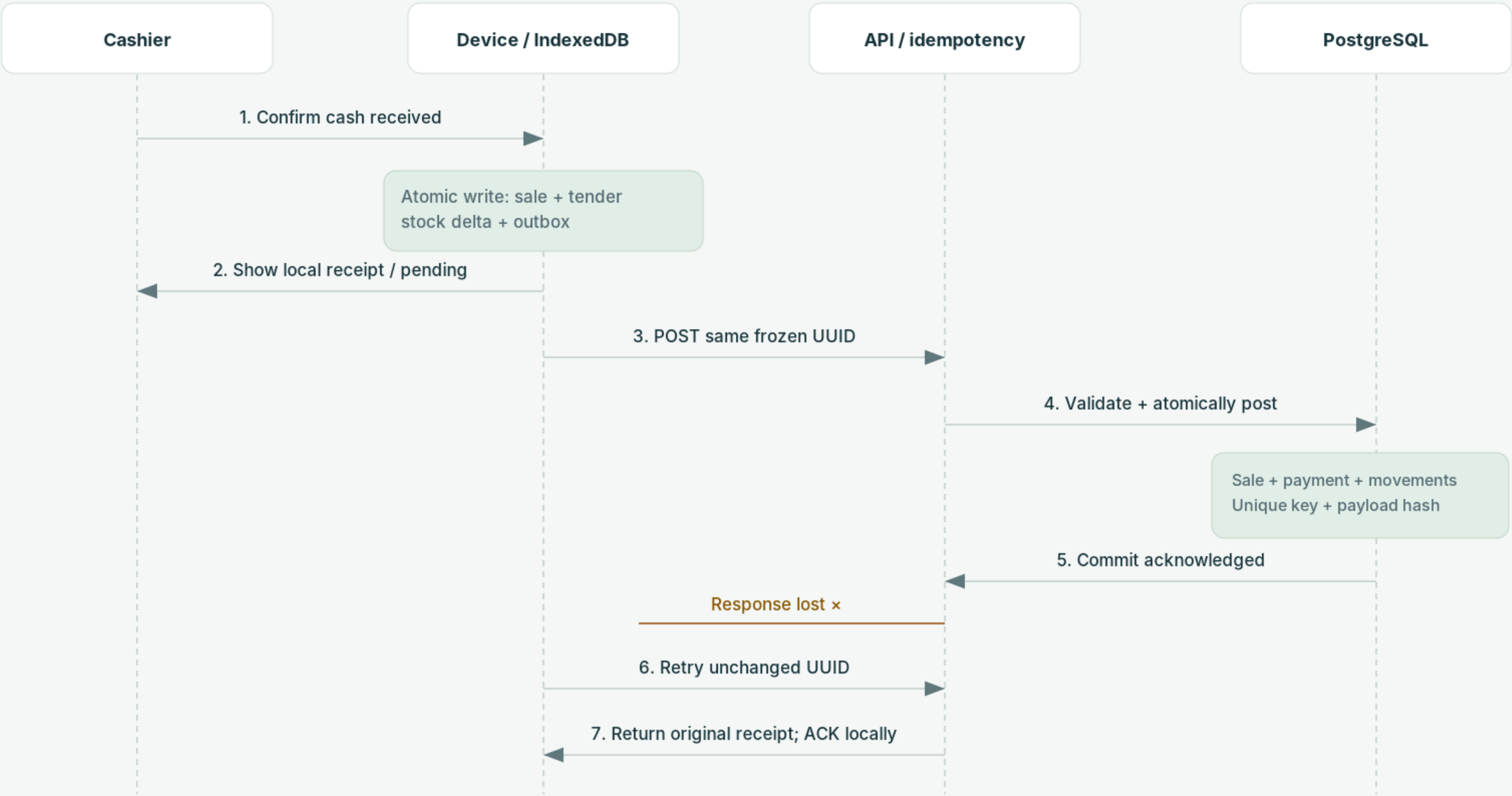
Engineering diagram / 03 sale sync sequence

Editable SVG and Mermaid source included. Read the full contract for exact constraints.

COUNTER POS / SYNC-01

A sale survives a lost response

Durable local commit first; an identical replay returns the original server result.



If content changes under the same UUID: 409 → NEEDS_REVIEW. Never invent a new UUID to “fix” a timeout.

Proposed V1 · Design contract, not a deployed application · 02 October 2026

Engineering diagram / 04 workflow states

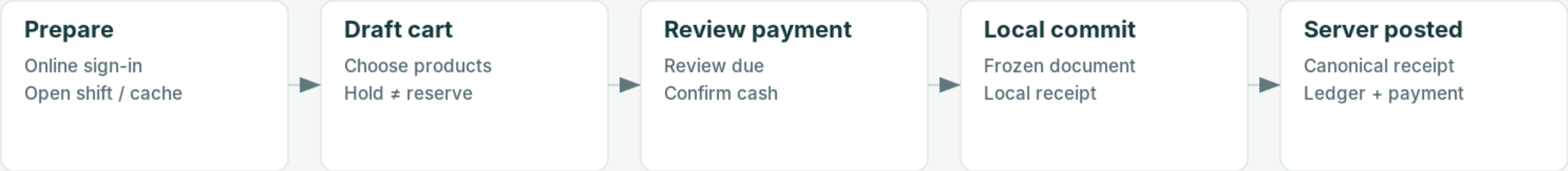
Editable SVG and Mermaid source included. Read the full contract for exact constraints.

COUNTER POS / FLOW-01

Business state is not synchronization state

A completed local sale is immutable even when server acknowledgement is pending.

BUSINESS WORKFLOW



OUTBOX LIFECYCLE

